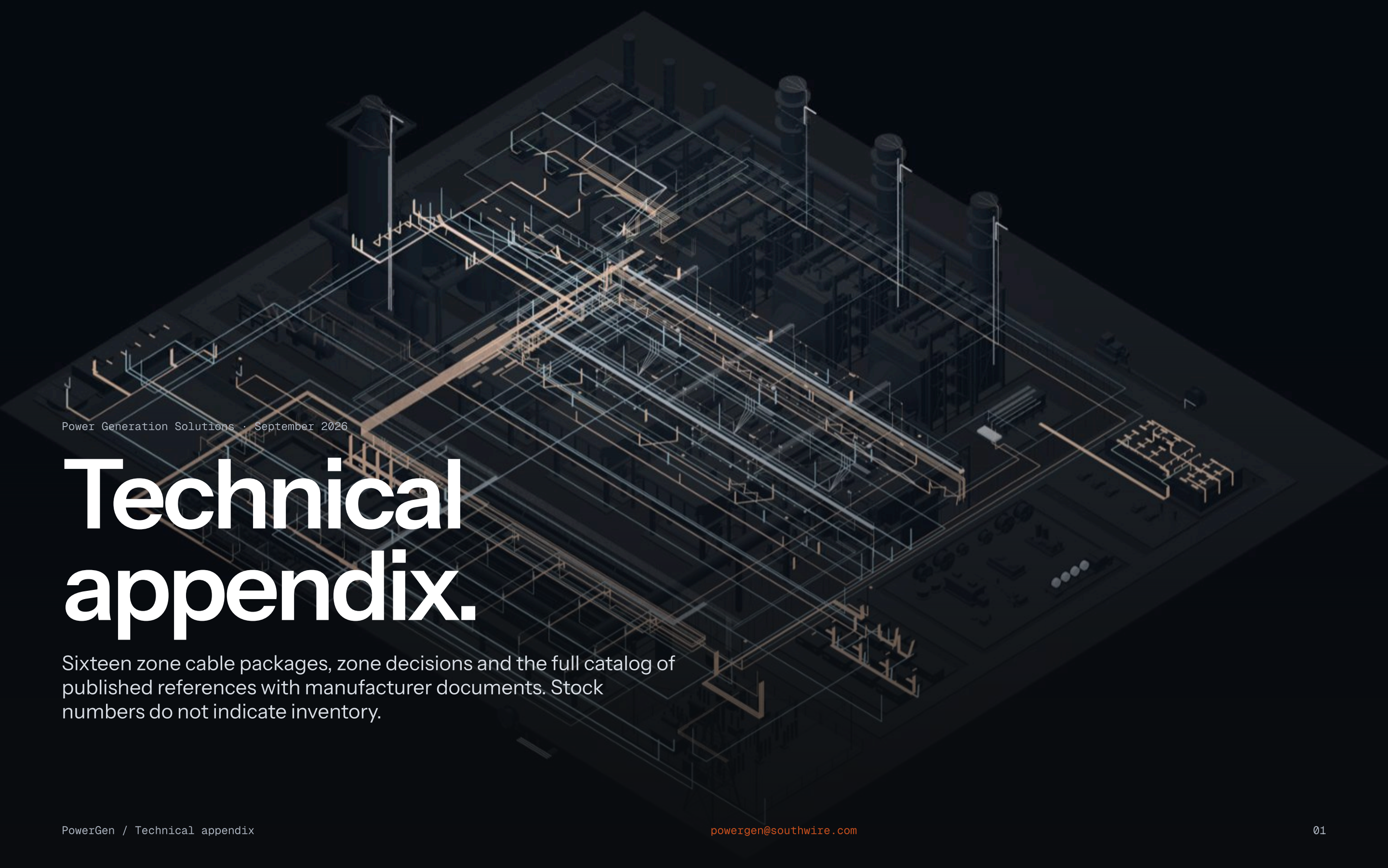




Power Generation Solutions · September 2026

Technical appendix.

Sixteen zone cable packages, zone decisions and the full catalog of published references with manufacturer documents. Stock numbers do not indicate inventory.

Applications 01–04. From equipment to scope.

01 Switchyard / GSU / grid tie

Station service and protection cable package

LV station service; CT/VT and breaker-control cable; trip/close DC; protection fiber; ground grid and equipment bonds.

Package focus: Define utility and GSU boundaries. HV underground cable is a separate engineered scope.

Specifies & buys: EPC / substation contractor via distribution; owner standards and utility team influence.

References [59563399](#) [59563699](#) [59547599](#) [59945899](#)

03 Reel staging / prefab

Cable delivery and assembly work packages

All destination-zone cable families; circuit labels, reel/cut lists, bundles, kits and installation documentation.

Package focus: Offer preterminated assemblies only to an agreed engineered scope. Staging adds no circuit footage.

Specifies & buys: EPC construction, electrical contractor and distributor; assembly scope with the qualified packager.

References [Services and staging scope](#)

02 Battery energy storage

Battery-to-collector interconnection package

RenewaFLEX for approved internal BESS wiring; MV collectors; PCS AC links; BMS/control, data and grounding.

Package focus: 13892403 is a 4/0, 2 kV BESS reference. 66131915 is a 35 kV EPR-insulated collector candidate only where the project design calls for it. Separate internal DC wiring from field AC scope.

Specifies & buys: BESS integrator owns internal design; EPC buys field AC, collector and site connections.

References [13892403](#) [66131915](#)

04 Admin / control room

Control-room power and building systems

MV/LV supply; UPS interfaces; SIMpull THHN/THWN-2 branch wire; applicable Genesis building systems; data and grounding.

Package focus: Select fire alarm, security and network listings by system; qualify plant OS2 fiber separately.

Specifies & buys: EPC / electrical contractor for feeds; controls integrator and low-voltage contractor for systems.

References [SIMpull Barrel THHN/THWN-2](#) [Genesis](#)

Click a stock reference for its technical document. Shared routes and staging do not add circuit footage.

Applications 05–08. From equipment to scope.

05 E-house

Primary distribution and factory interface package

5 kV or 15 kV-class MV source/tie feeders; Royal Guard LV secondaries; control, protection, DC, data and bonding.

Package focus: MV examples: 55052299 (5/8 kV) and 55050899 (15 kV). Confirm the one-line and separate OEM internal wiring from EPC field connections.

Specifies & buys: E-house / switchgear OEM for internal wiring; EPC / distributor for field feeders and ties.

References [55052299](#) [58950101 / 58950104](#) [55050899](#) [188040509](#)

07 Turbine hall

Turbine auxiliary and critical-power package

MV starting and pump feeds; LV lube-oil, turning-gear and vacuum loads; DLO emergency DC; control and temperature signals.

Package focus: Respect turbine OEM interfaces. Generator isolated-phase bus is separate.

Specifies & buys: Turbine OEM defines internal boundaries; EPC / electrical contractor buys external auxiliary circuits.

References [167022R108 / 167024R108](#) [59212899](#)

06 HRSG trains & stacks

Repeat train power and instrumentation package

MV pump and transformer feeds; LV motors; VFD outputs where fitted; control, shielded pairs, RTD and thermocouple extension.

Package focus: Package by train. EPR/CPE reference 59548199 contains four #16 triads; select the required sensor circuit count. Qualify hot-zone leads and heat tracing separately.

Specifies & buys: HRSG / auxiliary package OEM for internal scope; EPC electrical and I&C teams for field wiring.

References [59212899](#) [56295201](#) [59548199](#)

08 GT air inlets

Repeat inlet-house motor and sensor package

LV fan and pump feeders; drive output where fitted; damper control, differential-pressure signals, RTD and heating feeds.

Package focus: Confirm the modeled 3/C #2 feeder and motor count with the inlet OEM.

Specifies & buys: Turbine / inlet-house OEM and its packager; EPC for external supply connections.

References [Services and staging scope](#)

Click a stock reference for its technical document. Shared routes and staging do not add circuit footage.

Applications 09–12. From equipment to scope.

09 MCC / VFD / UPS

Motor-control and essential-power package

LV distribution; shielded VFD motor cable and terminations; UPS input/output and bypass; DLO battery feeds; control and bonding.

Package focus: Match output cable to the drive. Identify redundant AC and DC circuits.

Specifies & buys: MCC / VFD / UPS OEM for internal assembly; EPC and electrical contractor for field terminations.

References [64346799](#) [167022R108 / 167024R108](#) [59193199](#) [188040509](#)

11 Air-cooled condenser

Repeat fan-bank power and controls package

MV transformer feeds; 500 kcmil LV secondaries; 3/C 250 kcmil fan-bank feeders; control, temperature signals and bonding.

Package focus: Use 59199899 for the fixed feeder candidate; select VFD cable where the fan is drive-fed.

Specifies & buys: ACC OEM / fan-drive suppliers for package wiring; EPC for transformer and field motor feeds.

References [58950101 / 58950104](#) [59199899](#) [89052099](#)

10 Modular power yard

Repeat genset, black-start and fuel-cell interfaces

MV collectors; LV emergency and auxiliary power; OEM-approved starting-battery or fuel-cell DC; synchronization, control and data.

Package focus: Build repeat genset / RICE / fuel-cell interface kits from released drawings.

Specifies & buys: Genset / RICE / fuel-cell OEM and integrator for internal wiring; EPC for yard interconnects.

References [167022R108 / 167024R108](#)

12 Chillers / cooling towers

Cooling-system motor and control package

MV or LV pump and compressor feeders; fan/pump drive outputs; control, RTD, heater feeds and equipment bonds.

Package focus: Reconcile the cooling concept with the final motor list before multiplying packages.

Specifies & buys: Chiller / cooling tower OEM, mechanical packager and EPC electrical contractor.

References [64346799](#)

Click a stock reference for its technical document. Shared routes and staging do not add circuit footage.

Applications 13–16. From equipment to scope.

13 Water / wastewater

Pump, treatment skid and process-control package

MV pumps; LV RO, demineralization, chemical and effluent systems; drive output; level, flow, pressure and analyzer signals.

Package focus: Check chemical exposure and submersion. Dedicated pump leads may be needed.

Specifies & buys: Water treatment OEM / process skids; EPC electrical and I&C contractor for interconnects.

References [59212899](#)

15 Gas metering

Fuel-gas supply power and protection package

MV compressor supply; LV heaters and actuators; metering and shutdown signals; thermocouple/RTD; data and bonding.

Package focus: Consider ARMOR-X MC-HL where required. Match cable, glands and seals to area classification.

Specifies & buys: Fuel-gas skid / compressor supplier and EPC; process safety and electrical engineers set requirements.

References [89057289 / 89057399](#) [59563699](#) [89066999](#)

14 Carbon capture

Capture auxiliary power and instrumentation package

MV compressor/blower feeds; LV pumps; VFD outputs; control, shielded instrumentation, analyzers, data and bonding.

Package focus: Develop a separate CCS schedule. 15 kV ARMOR-X candidates include 89066999 and three-ground 45834799. Confirm motor voltage, drive requirements and the complete hazardous-location installation.

Specifies & buys: CCS technology provider / compressor and pump OEMs; EPC for field wiring and plant ties.

References [89057289 / 89057399](#) [89066999](#) [45834799](#)

16 Shared cable corridor

Plant-wide feeder and routing work packages

MV, LV, control, instrumentation, essential power, fiber and grounding routes connecting the plant zones.

Package focus: Create pull sections and reel plans. Count each circuit once. Select 5/8, 15 or 35 kV constructions from the actual system voltage and routing design.

Specifies & buys: EPC electrical / routing engineer, construction team and distributor; owner for resilience criteria.

References [55052299](#) [55050899](#) [89008499](#) [67747299](#)

Click a stock reference for its technical document. Shared routes and staging do not add circuit footage.

Specify the duty. Protect the interface.

Zones 06 / 07 / 08 • HRSG, turbine hall & air inlets

Heat, signals and OEM leads

Cables: MV auxiliaries, LV/VFD motors, RTD triads and type-matched thermocouple extension. Retain OEM probe, ignition and flame-scanner leads.

Before release: Confirm hot-route exposure, oil/vibration, shield treatment and entry space. Keep CEMS heated umbilicals and heat-trace elements in their vendor packages; define separate plant feeders.

Zones 14 / 15 • Capture & fuel systems

Account for the complete process

Cables: Compressor/pump MV, LV/VFD, MC-HL or TC-ER-HL where approved, analyzer/SIS signals, heat-trace feeders and cathodic-protection leads.

Before release: Add CCS power loads when process data is available. Obtain area classification, gas group, glands/seals and corrosion-control design. Hydrogen compatibility requires a defined system basis.

Zones 11 / 12 / 13 • Cooling & water

Wet service has different duties

Cables: Pump MV, fan/pump VFD outputs, actuator/control and analyzer pairs. Type W 44103 is an option for approved submerged pump-lead segments.

Before release: Distinguish washdown from continuous immersion; confirm chemicals, UV, starting voltage drop, drive lead length and motor termination. Specify true festoon/reeling duty separately.

Zones 04 / 05 / 09 • Control, e-house & MCC

Make power and controls fit

Cables: MV/LV feeders, VFD outputs, UPS/DC, relay control, instrument trunks, OS2 fiber and building-system cable.

Before release: Align cable OD, bend space, conductor class, lugs and terminal limits. Define CT/VT circuits, shield bonding and diverse protection paths. Use listed circuit-integrity systems only for compatible fire-alarm/communications duty.

Select constructions against actual routes and approved OEM data. The equipment shown does not establish cable size, quantity or a product guarantee.

Close the gaps between packages.

Zones 02 · Battery energy storage

Four systems within one fence

Cables: Approved field DC using RenewaFLEX or DLO; PCS AC; selected MV collection; and auxiliary, BMS, fire, fiber and grounding circuits.

Before release: Fix DC polarity/fault duty, connector and crimp compatibility, collector voltage and the factory-to-field boundary. Container count does not establish cable footage; do not add internal battery harnesses by default.

Zones 01 · GSU, switchyard & POI

One agreed grid handoff

Cables: Selected HV underground or overhead conductor; station service; battery trip/close; CT/VT, relay and SCADA circuits; ground grid and utility-approved fiber.

Before release: Confirm ownership, voltage, terminals, sheath bonding, ground-potential interfaces and diverse protection routes. Select OPGW or ADSS only when the overhead communications architecture requires it.

Zones 10 · Modular, RICE & standby power

Match the operating mission

Cables: Generator/collector feeders, radiators, jacket-water and lube pumps, starting DC, fuel controls and package networks.

Before release: Distinguish standby, emergency and true black-start duty. Define load steps, essential-bus/ATS interfaces and independent routes. Separate permanent feeders from temporary load-bank, welding and maintenance leads.

Zones 03 / 16 · Reels & shared corridors

Build the route into the package

Cables: Segregated MV/LV/control/fiber routes, ground continuity, penetrations, spare pathways and circuit-to-reel assignments.

Before release: Reconcile source-to-destination lengths, thermal conditions, fill, bends, pull direction, tension and sidewall pressure. Confirm end seals, access and firestop responsibility. Shared tray or duct length is not additional cable footage.

Select constructions against actual routes and approved OEM data. The equipment shown does not establish cable size, quantity or a product guarantee.

Medium-voltage power

Core references. Published references connect circuit requirements to specific constructions.

Stock / reference	Family	Duty	Construction / size	Spec	Documents	Zones
55052299	MV power	5/8 kV MV-105	750 kcmil Cu; NL-EPR, tape shield, CPE-TP	46103	Spec 46103 · PDF Product page	05, 16
55050899	MV power	15 kV MV-105	500 kcmil Cu; NL-EPR 220 mil, 133%, tape shield, CPE-TP	46304	Spec 46304 · PDF Product page	05, 16
89008499	MV power	35 kV MV-105	750 kcmil Cu; NL-EPR 420 mil, 133%, tape shield, SIMpull PVC	46702	Spec 46702 · PDF Product page	16
67747299	MV power	35 kV MV-105 aluminum	1250 kcmil AL compact; NL-EPR 420 mil, 133%, tape shield, SIMpull PVC	46710	Spec 46710 · PDF Product page	16
66131915 Collector candidate only where the project design calls for it. Product page, no spec PDF published in the source.	MV power	35 kV EPR concentric-neutral primary	1/C 1000 kcmil aluminum; EPR 420 mil, 133%; copper concentric neutral; polyethylene jacket	35 kV EPR	Spec 35 kV EPR · no PDF in source Product page	02
89066999	MV power MC-HL · hazardous location	ARMOR-X MC-HL 15 kV	3/C 500 kcmil Cu; 15 kV, 133%; NL-EPR; tape shields; 1 × #1 ground; ARMOR-X / red PVC	46454	Spec 46454 · PDF Product page	14, 15

Product family: selected by size and listing · specification number: a construction · published base code: not a complete orderable stock number · stock number: orderable, not inventory · request stock code: size and duty selected with the project.

Medium-voltage power

Expanded families. Published references connect circuit requirements to specific constructions.

Stock / reference	Family	Duty	Construction / size	Spec	Documents	Zones
890641 (base) 3/C 250 kcmil Published base code	MV power MC-HL · hazardous location	ARMOR-X 5/8 kV	3/C Cu NL-EPR, tape shields, armor/PVC; 5 kV 133% / 8 kV 100%. Match the complete MC-HL installation.	46255	Spec 46255 · PDF	05, 07, 09, 15
646579 (base) 1/C 500 kcmil Published base code	MV power	8 kV, 133% insulation	1/C NL-EPR / SIMpull PVC or 3/C ARMOR-X option. For an approved 6.9 kV basis requiring 133%; qualify drive duty.	46271 · 1/C	Spec 46271 · 1/C · PDF Spec 46284 · armor · PDF	05, 09, 11, 12
664985 (base) 3/C 500 kcmil Published base code Engineered alternate.	MV power MC-HL · hazardous location	ARMOR-X 25 kV	3/C Cu NL-EPR, 133%, tape shields, armor/PVC. Discrete generator-voltage alternative only after full system study; not a 35 kV collector cable.	46654	Spec 46654 · PDF	—
550783 (base) 1/C 500 kcmil Published base code	MV power	SOLONON LSZH MV	1/C Cu NL-EPR with LSZH-TP jacket; 5 kV 133% / 8 kV 100%. A smoke/halogen option; not a circuit-survivability rating.	46104	Spec 46104 · PDF	04, 05, 16
669087 (base) 3/C 500 kcmil Published base code	MV power	HaloFlex 3/C MV	Cu NL-EPR, tape shields and CPE-TP; 5 kV 133% / 8 kV 100%. MV-105 with CSA TC-ER marking; confirm the applicable installation approval.	46210	Spec 46210 · PDF	05, 07, 09, 16
Size / neutral selection Request stock code Request stock code Renewables and zone 02 option.	MV power	HI-DRI-PLUS collection	35 kV, 133% TRXLPE, moisture-blocked Al, full Cu concentric neutral and XLPE jacket. Confirm collector duty, neutral fault capacity and grounding.	80209	Spec 80209 · PDF	02

Product family: selected by size and listing · specification number: a construction · published base code: not a complete orderable stock number · stock number: orderable, not inventory · request stock code: size and duty selected with the project.

Low-voltage, VFD and flexible / DC

Core references. Published references connect circuit requirements to specific constructions.

Stock / reference	Family	Duty	Construction / size	Spec	Documents	Zones
45834799	VFD MC-HL · hazardous location	ARMOR-X MC-HL 15 kV VFD	3/C 350 kcmil Cu; 15 kV, 133%; NL-EPR; 3 × #6 grounds; ARMOR-X / red PVC; qualify drive system	46456	Spec 46456 · PDF Product page	14
58950101 / 58950104	LV power	Royal Guard 600 V	500 kcmil; EPR / thermoset CPE; 1,000 / 2,000 ft	45005	Spec 45005 · PDF Product page	05, 11
58949999	LV power	Royal Guard 600 V	350 kcmil; LV distribution single	45005	Spec 45005 · PDF	—
59199899	LV power	TC-ER motor feeder	3/C 250 kcmil + #4 ground; EPR XHHW-2 / CPE-TP	45255	Spec 45255 · PDF Product page	11
64346799	VFD	Machine Flex VFD	4 AWG representative; shielded XLPE / TPE	45440	Spec 45440 · PDF	09, 12
89052099	VFD MC-HL · hazardous location	ARMOR-X VFD	3/C 250 kcmil + 3 × #8 grounds; XLPE / PVC; MC-HL	45351	Spec 45351 · PDF Product page	11
167022R108 / 167024R108	Flexible / DC	DLO flexible power	1/C 2/0 or 4/0; 2 kV; EPDM / thermoset CPE	44020	Spec 44020 · PDF Product page	07, 09, 10
13892403 Approved internal BESS wiring. Separate internal DC wiring from field AC scope; not a solar PV application.	Flexible / DC	RenewaFLEX BESS	4/0 Cu; RHH/RHW-2, 2 kV; orange; 500 ft	44080	Spec 44080 · PDF Product page	02

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Low-voltage, VFD and flexible / DC

Expanded families. Published references connect circuit requirements to specific constructions.

Stock / reference	Family	Duty	Construction / size	Spec	Documents	Zones
11308401 · Cu 500 / 11277115 · Al 4/0 LV feeders and buildings.	LV power	SIMpull XHHW-2	Cu or AlumaFlex aluminum XLPE singles; select permitted size/rating and installation. Aluminum requires compatible terminals and thermal review.	10006 · Cu	Spec 10006 · Cu · PDF Spec 10015 · Al · PDF	—
Size to drive duty Request stock code Request stock code	VFD	Machine Flex VFD	1000 V XLPE/TPE; shielded drive-output construction with symmetrical grounds. Confirm lead length, motor/drive and termination system.	45443	Spec 45443 · PDF	09, 11, 12, 13
558157 (base) 4/C #2 Published base code	Flexible / DC	Type W / submerged lead	4/C, 2 kV EPDM/CPE. Approved portable or continuously submerged pump-lead duty; not general permanent building wiring.	44103	Spec 44103 · PDF	10, 13
Duty / size selection Request stock code Request stock code Maintenance and portable use.	Flexible / DC	Royal SOOW	600 V flexible EPDM/CPE cord for extra-hard usage. Select conductor count, oil/water exposure and approved connection; separate from fixed plant takeoff.	—	Product page	—
Size / core selection Request stock code Request stock code Building and auxiliary feeders.	LV power	Jacketed Armorlite MC	600 V Cu THHN/THWN singles, ground, interlocked armor and PVC jacket. Coordinate wet-location suitability and fittings; distinct from MC-HL.	60135	Spec 60135 · PDF	—
Size / terminal selection Request stock code Request stock code Solar and hybrid projects.	Flexible / DC	2 kV Type PV	Cu or aluminum XLPE single conductor, RHH/RHW-2 PV. Solar DC application with matched connectors; not an automatic BESS cable substitute.	45411 · Cu	Spec 45411 · Cu · PDF Spec 45410 · Al · PDF	—
89052799 4/C #8 + #10 ground	LV power MC-HL · hazardous location	ARMOR-X LV power	600/1000 V, 4/C Cu XLPE XHHW-2, ground, welded armor and PVC jacket. MC-HL power option; coordinate area classification and fittings.	45352	Spec 45352 · PDF	05, 09, 14, 15
55085699 4/C 4/0 + #4 ground	LV power	ARMOR-X LSZH power	600/1000 V, 4/C Cu XLPE, ground, armor and SOLONON LSZH-TP jacket. Select for the smoke/halogen basis; it is not a fire-survival rating.	45354	Spec 45354 · PDF	04, 05, 09
Product family: selected by size and listing · specification number: a construction · published base code: not a complete orderable stock number · stock number: orderable, not inventory · request stock code: size and duty selected with the project.						

Control, instrumentation and protection

Core references. Published references connect circuit requirements to specific constructions.

Stock / reference	Family	Duty	Construction / size	Spec	Documents	Zones
59563399	Control & I	CT/VT control	4/C #10; tinned Cu; EPR XHHW-2 / CPE	45059	Spec 45059 · PDF Product page	01
59563699	Control & I	CT/VT control	4/C #12; tinned Cu; EPR XHHW-2 / CPE	45059	Spec 45059 · PDF Product page	01, 15
59547599	Control & I	Breaker control	19/C #14; EPR / CPE	45059	Spec 45059 · PDF Product page	01
59193199	Control & I	Output / control	2/C #16; unshielded EPR XHHW-2 / CPE; Type TC	45059	Spec 45059 · PDF Product page	09
577883 (base code) Published base code Published base code, not a complete orderable stock number.	Control & I	Shielded output / control	2/C #16; foil shield; PVC/nylon / PVC	45154	Spec 45154 · PDF	—
59212899	Control & I	Analog / status pair	1 pair #18; overall shield; EPR / CPE	43618	Spec 43618 · PDF	06, 07, 13
56295201	Control & I	RTD triad candidate	1 triad #18; overall shield; PVC/nylon / PVC	43606	Spec 43606 · PDF	06
89057289 / 89057399	Control & I MC-HL · hazardous location	ARMOR-X instrumentation	2 / 4 pairs #18; armored MC-HL construction	43702	Spec 43702 · PDF	14, 15
59945899	Control & I	Substation control	4/C #14 tinned Cu; EPR XHHW-2 / CPE-TP; TC-ER; black/white/red/green	45059	Spec 45059 · PDF Product page	01
59548199 Four #16 triads. Match sensor count and jacket to the specification.	Control & I	EPR/CPE instrumentation triads	4 triads, #16 tinned Cu; individual + overall shields; EPR / CPE; TC-ER	43624	Spec 43624 · PDF Product page	06

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Control, instrumentation and protection

Expanded families. Published references connect circuit requirements to specific constructions.

Stock / reference	Family	Duty	Construction / size	Spec	Documents	Zones
582649 (base) 5/C #16 Published base code	Control & I	Machine Flex control	Flexible Cu PVC/nylon/TPE; 600/1000 V. Select conductor count and applicable TC-ER rating; 75 °C wet / 90 °C dry.	45041	Spec 45041 · PDF	05, 07, 10
679792 (base) 7/C #14 Published base code	Control & I MC-HL · hazardous location	Halo-Flex control	XLPE/CPE-TP flexible Cu; 600 V TC-ER-HL or 1000 V TC-ER. Confirm area, gland, seal and full-system approval.	45061	Spec 45061 · PDF	14, 15
679955 (base) 2 pairs #16 Published base code	Control & I MC-HL · hazardous location	Halo-Flex instrument	600 V PVC/nylon pairs, individual and overall shields, CPE jacket. Select approved TC-ER-HL construction and shield treatment.	43632	Spec 43632 · PDF	14, 15
Alloy / size selection Request stock code Request stock code	Control & I	Thermocouple extension	300 V PVC/PVC or 600 V EPR/CPE multipair. Match extension alloy, polarity, shields and temperature; ordinary copper is not equivalent.	42350 · 300 V	Spec 42350 · 300 V · PDF Spec 42355 · 600 V · PDF	06, 07, 14
596821 (base) 1 pair #16 Published base code Process analog and status.	Control & I	300 V PLTC / ITC	PVC/PVC pairs with overall shield. Select pair count, circuit classification and routing; not interchangeable with power cable.	43301	Spec 43301 · PDF	—
589289 (base) 4/C #14 Published base code	Control & I MC-HL · hazardous location	ARMOR-X control	600 V Cu XLPE XHHW-2, insulated ground, welded armor and PVC jacket; MC-HL. Control construction is distinct from shielded instrumentation.	45227	Spec 45227 · PDF	05, 09, 14, 15
58142502 · PVC 4 pairs #16, 2,000 ft	Control & I	600 V instrument trunks	Individually shielded pairs plus overall shield: PVC/nylon-PVC or EPR/CPE. Confirm item listing and shield scheme; avoid duplicating field pairs.	43604 · PVC	Spec 43604 · PVC · PDF Spec 43620 · EPR · PDF	04, 06, 07, 14

Product family: selected by size and listing · specification number: a construction · published base code: not a complete orderable stock number · stock number: orderable, not inventory · request stock code: size and duty selected with the project.

Grounding, networks, building systems and specialty

Core references. Published references connect circuit requirements to specific constructions.

Stock / reference	Family	Duty	Construction / size	Spec	Documents	Zones
188040509	Specialty	SIS switchboard wire	4 AWG 133-strand tinned Cu; XLPE; 600 V, 90 °C; switchgear and panel internal wiring	LW93	Spec LW93 · PDF Product page	05, 09
10704542 / 16266909	Grounding	Bare copper ground	4/0 7-strand and 2/0 37-strand published examples; final sizes follow the grounding study	10080	Spec 10080 · PDF	01, 16
SIMpull Barrel THHN/THWN-2 Product family Family reference; select the stock number by size.	Specialty	Building branch wire	THHN/THWN-2 branch wire for building systems	10026	Spec 10026 · PDF	04
Genesis Product family Product family page; no single stock number applies.	Specialty	Fire alarm, security and building-control cable	Select the actual system listing	—	Product page	04
VFD spec sheets 85451 / 85452 / 85456 Product family	Specialty	Terminations and accessories	MV terminations, lugs, glands, VFD shield connections, identification and supports	85451 / 85452 / 85456	Spec 85451 / 85452 / 85456 · PDF Spec 85452 · PDF Spec 85456 · PDF	—

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Grounding, networks, building systems and specialty

Expanded families. Published references connect circuit requirements to specific constructions.

Stock / reference	Family	Duty	Construction / size	Spec	Documents	Zones
109298 (base) #2 Cu, 7-strand Published base code	Specialty	Cathodic-protection lead	Copper with LLDPE/HMWPE covering. Separate corrosion-control takeoff; covering is not rated insulation. Confirm actual service and isolation joints.	83350	Spec 83350 · PDF	13, 14, 15
SC4001-2F 2/C #18 unshielded Alarm and communications only; not motor or fire-pump power cable.	Specialty	Circuit Defender	2-hour listed system for compatible low-voltage circuits. Family is 300 V; fire-resistive use follows System 44, including its 72 V limit.	13001 · plain	Spec 13001 · plain · PDF Spec 13002 · shield · PDF	—
I99981-1A Published item	Specialty	Genesis Cat 6A CMP	23 AWG, 4-pair U/UTP plenum example, blue, 1,000 ft reel. Unshielded; exposed plant, Ethernet/fieldbus and fiber links need their own ratings.	13140	Spec 13140 · PDF Product page	04

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Additional cases with a reference.

Onshore HV connection / Vineyard Wind 1

32 miles

High-voltage cable, manufactured and installed.

Problem. An onshore high-voltage cable system for a generation project's grid connection.

Contribution. Design, manufacturing and installation of the project's onshore high-voltage cable system.

Result. Southwire's case study describes design, manufacturing and installation of the project's onshore high-voltage cable system.

PowerGen use, proposed. Engage the HV team early when a generation project requires a qualified underground gen-tie or grid connection.

[Case study · PDF](#)

URD reliability program / NV Energy

\$800 million

Replacement cost deferred.

Problem. A large population of aging underground distribution cable.

Contribution. A multi-year rejuvenation program described in a Southwire-hosted paper.

Result. The paper reports over 12 million conductor feet rejuvenated through 2020 and the utility's estimate of more than \$800 million in deferred replacement cost.

PowerGen use, proposed. Use condition assessment to compare rejuvenation, repair and replacement for eligible aging plant feeders. Utility program results are not a forecast for a generation site.

[Case study · PDF](#)

69 kV transmission / Atlantic Municipal Utilities

1969 vintage

A 69 kV underground circuit kept in service.

Problem. A 1,300-foot underground 69 kV section originally installed in 1969.

Contribution. Rejuvenation of the existing 500 kcmil aluminum, 69 kV XLPE cable.

Result. Southwire's paper describes rejuvenation of the section rather than replacement.

PowerGen use, proposed. For suitable brownfield circuits, evaluate cable condition, accessories and outage requirements before choosing rejuvenation or replacement.

[Case study · PDF](#)